

EXAMINATION: ELECTRONIC COMPONENT MANAGEMENT SYSTEM (ECMS)

1. General Requirements

- **Base URL:** <http://localhost:8084/ElectronicSystem/>
- **Tech Stack:** Java Servlet, JSP, JDBC, SQL Server
- **Database:** ComponentDB

2. Database Schema & Sample Data

Use the following structure to initialize your database:

```
CREATE TABLE tblUsers (  
    userID varchar(20) PRIMARY KEY,  
    fullName nvarchar(50) NOT NULL,  
    password varchar(50) NOT NULL,  
    roleID varchar(5), -- 'AD': Admin, 'ST': Staff , 'CS':Customer  
    status int      -- 1: Active, 0: Inactive [cite: 20]  
);
```

```
CREATE TABLE tblComponents (  
    itemID varchar(10) PRIMARY KEY,  
    itemName nvarchar(100) NOT NULL,  
    category nvarchar(50),  
    manufacturer nvarchar(50),  
    specification nvarchar(255),  
    warehouseLocation varchar(20),  
    importDate date,  
    quantity int,  
    unitPrice money,  
    warrantyMonth int,  
    isAvailable bit, -- 1: Available, 0: Discontinued [cite: 31, 36]  
    note nvarchar(255)
```

);

```
INSERT INTO tblUsers (userID, fullName, password, roleID, status) VALUES
('admin01', 'System Administrator', 'admin@123', 'AD', 1),
('admin02', 'Nguyen Admin', 'password', 'AD', 1),
('staff01', 'Technical Support', 'staff@123', 'ST', 1),
('cust01', 'Warehouse Keeper', '123456', 'CS', 1),
('baduser', 'Terminated Staff', '1', 'ST', 0), [cite: 42]
('locked_ad', 'Inactive Manager', '1', 'AD', 0);
```

```
INSERT INTO tblComponents VALUES
```

```
-- Available items (Should appear in Search)
```

```
('C001', 'ESP32 WROOM', 'MCU', 'Espressif', 'WiFi+BT', 'Shelf A1', '2023-01-05', 50,
150000, 12, 1, 'New'), [cite: 44]
('C002', 'LM7805', 'IC', 'TI', '5V Regulator', 'Shelf B2', '2023-02-10', 200, 5000, 6, 1,
'Bulk'), [cite: 45, 48, 52]
('C003', 'OLED 0.96', 'Display', 'Generic', 'I2C Blue', 'Shelf C1', '2023-03-15', 30,
85000, 3, 1, 'Fragile'), [cite: 53, 54]
('C005', 'Resistor 10k', 'Passive', 'Yageo', '1/4W Carbon', 'Box 01', '2023-01-20', 1000,
200, 0, 1, 'Small'), [cite: 57, 58]
('C006', 'SG90 Servo', 'Motor', 'TowerPro', '9g Micro', 'Shelf A5', '2023-05-12', 15,
45000, 6, 1, 'Popular'), [cite: 59, 61]
('C009', 'L298N', 'Driver', 'ST', 'Dual H-Bridge', 'Shelf B3', '2023-06-01', 25, 65000, 6,
1, 'High power'), [cite: 63, 64]
('C011', 'Capacitor 100uF', 'Passive', 'Rubycon', '25V Electrolytic', 'Box 02', '2024-01-
10', 500, 1500, 0, 1, 'Standard'),
('C012', 'HC-SR04', 'Sensor', 'Elecrow', 'Ultrasonic Distance', 'Shelf D1', '2024-02-14',
45, 25000, 6, 1, 'In stock'),
('C004', 'DHT11', 'Sensor', 'Aosong', 'Temp/Humid', 'Shelf D4', '2023-04-20', 0, 35000,
1, 1, 'Out of stock'), [cite: 55, 56]
('C007', 'Uno R3', 'Board', 'Arduino', 'ATmega328P', 'Shelf A1', '2023-02-25', 10,
250000, 12, 0, 'Discontinued'), [cite: 62]
('C008', 'LED Red 5mm', 'Diode', 'Generic', 'Standard Red', 'Box 05', '2023-01-01',
500, 500, 0, 0, 'Clearance'), [cite: 62]
```

('C010', 'Buzzer 5V', 'Audio', 'Generic', 'Active Piezo', 'Box 08', '2023-03-10', 5, 0, 0, 1, 'Free Sample'); [cite: 65, 66]

3. Full Functional Specifications

Function 1: Secure Login

- **Input Parameters:** userID, password.
- **Error Handling Logic:**
 - **Invalid ID:** Display "User does not exist" if the ID is not in the database.
 - **Invalid Password:** Display "Incorrect Password" if credentials do not match.
 - **Inactive Account:** Display in another page "This account is not activated" if status = 0.
- **Successful Login:** Store the User object in the **Session** and redirect to home.jsp.

Function 2: Header & Authorization

- **Welcome Message:** Display "Welcome [fullName] ([roleID])" prominently on every page.
- **Navigation:** home.jsp must provide clear links to the **Search Page** and **Logout**.
- **Logout Logic:** Invalidate the current Session and redirect the user back to login.jsp.

Function 3: Search Components

- **Entry Point:** componentList.jsp.
- **Behavior:**
 - If the search input is **empty**, the data table must remain hidden.
 - Search by itemName and category
 - **Filters:** Do not display items where unitPrice <= 0 or isAvailable = 0 or quantity <= 0.
 - If Admin search, show component (ignore filter condition)
 - **Output:** Display 10 columns of information (all fields from tblComponents except isAvailable).

Function 4: Component Update (Admin Only)

- **Interface:** An "Edit" link/button appears in the search result table **only** if the user is an Admin (roleID = 'AD' or 'ST').
- **Workflow:**
 - Clicking "Edit" fetches the specific component details and populates an update.jsp form.
 - **Fields for modification(Admin):** itemName, category, manufacturer, specification, warehouseLocation, quantity, unitPrice, warrantyMonth, isAvailable, note.

- **Fields for modification(Admin):** itemName, category, manufacturer, specification, warehouseLocation, quantity, unitPrice, warrantyMonth, isAvailable, note.
- **Fields for modification(Staff):** itemName, category, manufacturer, specification, warehouseLocation, quantity, note. (can't change status isAvailable)
- IsAvailable must be a checkbox.
- Auto update importDate to the date updated.(both Staff and Admin)
- **Constraints:** quantity must be quantity and unitPrice must be greater than 0.
- **Post-Update:** On success, redirect to the Search page while **preserving the previous search keyword**.

Function 5: System Security & Access Control

- **Role Restriction:** The Update action must be protected at the Controller/Servlet level.
- **Unauthorized Access:** If a Staff member (ST) attempts to access the Update URL directly:
 - Block the transaction.
 - Redirect to componentList.jsp.
 - Display a red error message: **"No Permission: Admin access required"**.